



National Institute of Technology Kurukshetra
Department of Computer Engineering
Digital System Design (CSIC100)(IT, AIML)
MID-II, APR-2025

Timing: 50 mins

Max Marks: 15

All the Questions are Compulsory.

1. (a) Calculate the minimum required two-input NAND Gates to Implement the Boolean expression $Y = \bar{A}BC + \bar{A}\bar{B}\bar{C} + A\bar{B}\bar{C} + A\bar{B}C$. Draw the logic diagram. (3)
- (b) Demonstrate the validity of the following identities by means of truth tables: (3)

- $(x + y + z)' = x'y'z'$

- $(xyz)' = x' + y' + z'$

2. (a) Implement the following Boolean function F , using the two-level forms of logic NAND-AND gates. (3)

$$F(A, B, C, D) = \sum(0, 4, 8, 9, 10, 11, 12, 14)$$

- (b) Using Quine-Mccluskey(QM) Tabular Method to simplify the following Boolean Expression. (3)

$$F(W, X, Y, Z) = \sum(0, 4, 5, 7, 8, 11, 12, 15)$$

3. Implement the following three boolean expressions using three half adders. (3)

$$D = A \oplus B \oplus C$$

$$E = A'BC + AB'C$$

$$F = ABC' + (A' + B')C$$